FIN2020 Homework 2

Due **5pm Wednesday Feb 4**

1. Expected Utility and Risk Aversion

Consider an investor with von Neumann-Morgenstern expected utility function

$$U(x, y, \pi) = \pi u(W_0 + x) + (1 - \pi)u(W_0 + y),$$

defined over lotteries that offer payoff x with probability π and payoff y with probability $1 - \pi$, where W_0 measures the investors wealth before accepting the lottery. As we discussed in class, concavity of the Bernoulli utility function u captures the investor's aversion to risk.

To verify this with an example, suppose that the Bernoulli utility function takes the specific form

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

so that

$$u'(c) = c^{-\gamma}$$

and

$$u''(c) = -\gamma c^{-\gamma-1}.$$

This last expression for $u''(c)$ shows that the investor will be risk averse if the preference parameter γ is positive and, will become in some sense, "more risk averse" the larger is γ.

Suppose that $W_0 = 10$, so that the investor starts out with \$10, and consider the following three lotteries:

1. Lottery one has $(x, y, \pi) = (5, 0, 1/2)$, so that the investor wins \$5 with probability 1/2 and gets nothing with probability 1/2.
2. Lottery two has $(x, y, \pi) = (2.5, 0, 1)$, so that the investor gets \$2.50 with probability one, that is, for sure.
3. Lottery three has $(x, y, \pi) = (2, 0, 1)$, so that the investor gets \$2 with probability one, that is, for sure.

Using these numbers, compute and compare the values of

$$U(x, y, \pi) = \pi u(W_0 + x) + (1 - \pi)u(W_0 + y) = \pi \left[\frac{(W_0 + x)^{1-\gamma}}{1 - \gamma} \right] + (1 - \pi) \left[\frac{(W_0 + y)^{1-\gamma}}{1 - \gamma} \right]$$

for all three lotteries when the investor has $\gamma = 1/2$. Then re-do the exercise with $\gamma = 2$ and $\gamma = 3$. Looking across the results for all values of γ, does the investor ever prefer the risky bet of \$5 versus 0 to the safe option that offers the average of \$2.50 for sure? Why or why not? Then compare the risky bet of \$5 versus 0 to the safe option that offers only \$2 for sure. For which value(s) of γ does the investor prefer the risky bet to the safe choice of \$2 for sure? For which value(s) of γ does the investor prefer the safe choice of \$2 to the risky bet? For which value(s) of γ is the investor indifferent between the risky bet and \$2 for sure?

2. Interpreting Measures of Risk Aversion

Consider an investor who has vN-M expected utility with Bernoulli utility function

$$u(Y) = \frac{Y^{1-\gamma} - 1}{1 - \gamma},$$

where, as we now know, $\gamma > 0$ represents the coefficient of relative risk aversion.

Suppose that the investor's initial wealth is $Y_0 = 10$ and that he or she is confronted with the lottery $(0.1, -0.1, \pi)$. For values of γ equal to 1/2, 2, 3, 10, and 20, compute the exact value of π^* that makes the investor indifferent between accepting or rejecting the bet. Recall from class that this value of π^* can be approximated by

$$\pi^* \approx \frac{1}{2} + \frac{1}{4} \left[-\frac{Y u''(Y)}{u'(Y)} \right] k$$

where k is the size of the bet as a fraction of initial wealth: 0.01, or one percent, in this case. Compare your exact answers to the values implied by the formula to see how well the approximation works.